Stable Quantum Ghost

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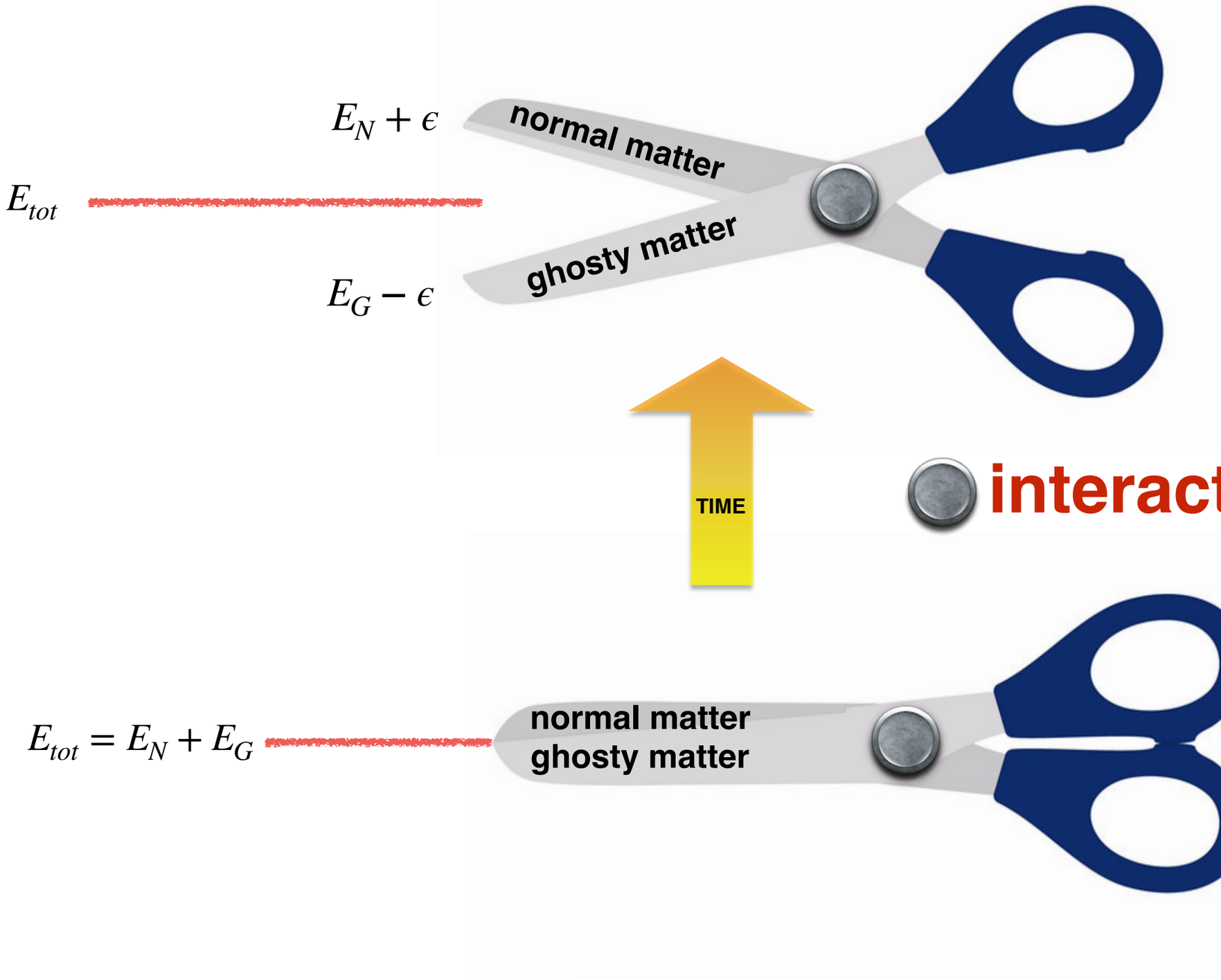
Interacting Ghost with Pure Point Spectrum

Ghosts-
are dynamical degrees of freedom
with negative mass -
i.e. with kinetic energy unbounded from below

Ghosts
are fields propagating in an effective metric with signature
opposite to the one chosen for
usual healthy fields

Sawicki, Trenkler, Vikman (2024)

Runaway Instability



Ghosts instabilities

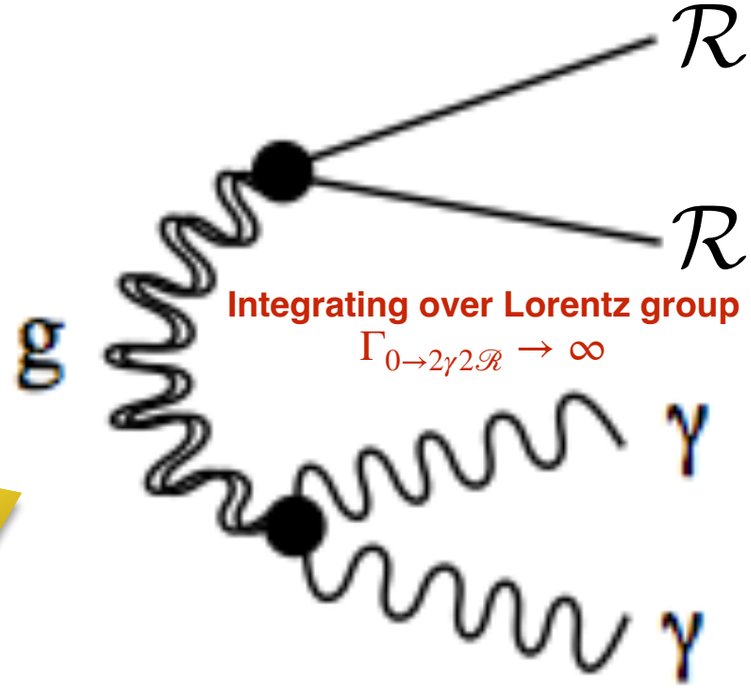
$$S[\mathcal{R}] = \frac{1}{2}\int d\tau d^3\mathbf{x}\, Z\left(\left(\mathcal{R}'\right)^2 - c_{\rm S}^2\left(\partial_i\mathcal{R}\right)^2\right)$$

$$H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

$$H_{\mathbf{k}} = \frac{\left|P_{\mathbf{k}}\right|^2}{2Z} + \frac{Z c_S^2 k^2 \left|\mathcal{R}_{\mathbf{k}}\right|^2}{2}$$

ghost $Z(t) < 0$

- *oscillators with* ***negative mass,*** $m < 0$
- ***"wrong" signature convention***



$$\Gamma_{0\to 2\gamma 2\mathscr{R}} \sim \Lambda^8/M_{Pl}^4$$

Cline, Jeon, Moore,(2003)

Why are we interested in ghosts?

- Interesting cosmology: Phantom Dark Energy (DESI recent results) - super accelerated universe with $\dot{H} > 0$, Bouncing universe etc. Inflation with blue spectrum of gravity waves

- Higher-Derivative systems have ghosts due to the "Ostrogradsky theorem" (1848), see nice reviews by Woodard (2006), (2015) and his talk from last week

- Renormalization / Quantization of Gravity (Stelle, 1977)

$$S = \int R + \alpha R^2 + \beta W_{\mu\nu\sigma\lambda} W^{\mu\nu\sigma\lambda}, \quad \text{Weyl tensor} \quad W = \partial\partial g,$$

  We kind of see R^2 "Starobinsky inflation". Where is the Weyl term?

- Questions related to entropy and thermodynamics

- Is it possible to screen gravity?

- Is it possible to screen the Cosmological Constant or the energy of quantum vacuum?

- Interesting exotic compact objects like wormholes

- Can gravitons be massive? (Boulware–Deser ghost, 1972, dRGT etc.)

Let us try to understand first
what happens
with a finite number of degrees of freedom

classical Local and *Global* Stability is proven for some mechanical models in

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Ghosts without Runaway Instabilities

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We present a simple class of mechanical models where a canonical degree of freedom interacts with another one with a negative kinetic term, i.e., with a ghost. We prove analytically that the classical motion of the system is completely stable for all initial conditions, notwithstanding that the conserved Hamiltonian is unbounded from below and above. This is fully supported by numerical computations. Systems with negative kinetic terms often appear in modern cosmology, quantum gravity, and high energy physics and are usually deemed as unstable. Our result demonstrates that for mechanical systems this common lore can be too naive and that living with ghosts can be stable.

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Global and local stability for ghosts coupled to positive energy degrees of freedom

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Unitary Time Evolution and Vacuum for a Quantum Stable Ghost

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We quantize a classically stable system of a harmonic oscillator polynomially coupled to a ghost with negative kinetic energy. We prove that due to an integral of motion with a positive discrete spectrum: i) the Hamiltonian has a pure point spectrum unbounded in both directions, ii) the evolution is manifestly unitary, iii) the vacuum is well-defined, iv) expectation values for squares of canonical variables are bounded. Numerical solutions of the Schrödinger equation confirm these results. We argue that the discrete spectrum of the integral of motion enforces stability for extended interactions.

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Now QM of these models

e-Print: 2604.21826 [hep-th]

Quantum mechanics with a ghost: Counterexamples to spectral denseness

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We quantise integrable point-particle systems with opposite-sign kinetic terms and nontrivial interactions. Using methods from separability theory, we show that previously determined classical stability conditions also imply discrete separated eigenvalue spectra. The resulting energy spectrum is unbounded above and below but not necessarily dense. We establish sufficient conditions for (i) exactly one accumulation point, or (ii) none at all. This dispels the widespread notion that ghostly quantum systems must have a continuous or dense energy spectrum.

ANNALS OF PHYSICS 146, 209–220 (1983)

Some Quantum Operators with Discrete Spectrum but Classically Continuous Spectrum

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Received September 22, 1982

We consider a number of simple quantum Hamiltonians $H(-i\nabla, x)$ with the following property: $H(-i\nabla, x)$ has discrete spectrum even though $\{(p, q) \mid H(p, q) < E\}$ has infinite volume.

1. INTRODUCTION

A standard rule of thumb about whether a quantum Hamiltonian $H = -\Delta + V(x)$ has purely discrete spectrum or some continuous spectrum is the following. Look at the volume $\{(p, q) \mid p^2 + V(q) \leqslant E\}$ in $R^{2\nu}$. If this volume is finite for all E, the standard wisdom is that H has only discrete spectrum. This standard wisdom also says that if this volume is infinite for some $E < \infty$, then the spectrum is not purely discrete. There are theorems which validate the first half of this intuition.

System to Quantize

system:

$$H = \frac{p_x^2 + x^2}{2} - \frac{p_y^2 + \omega^2 y^2}{2} + V_I(x, y)$$

for global stability $c > 0, \gamma > 0$

$$V_I = \frac{1}{2c}(1 - \omega^2)(x^2 - y^2)^2 + \gamma\left[(x^2 - y^2)^3 + c(x^4 - y^4)\right]$$

to decouple $c \to \infty, \gamma \to 0$

Integral of Motion:

$$H_\uparrow = p_x^2 + p_y^2 + \frac{2}{c}K^2 + V_\uparrow(x, y) \qquad K = x\,p_y + y\,p_x$$

$$V_\uparrow = x^2 + \omega^2 y^2 + \frac{1}{c}\left(1 - \omega^2\right)\left(x^4 - y^4\right) + 2\gamma\left(\left(x^2 - y^2\right)^2\left(x^2 + y^2\right) + c\left(x^4 + y^4\right)\right)$$

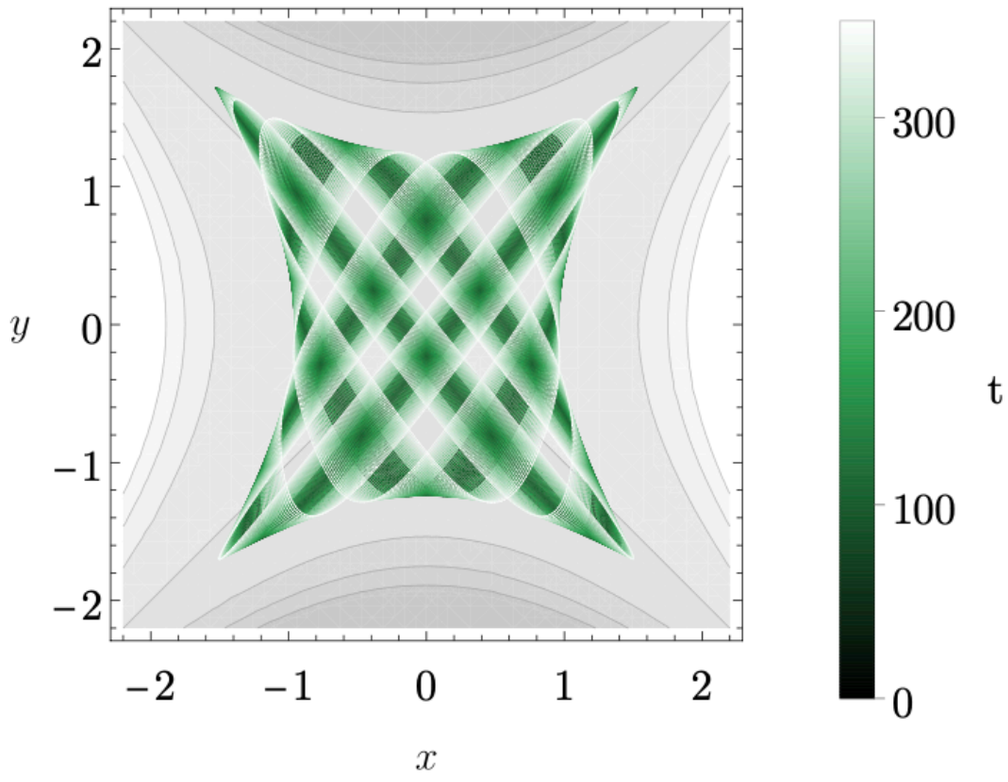
binding the motion:

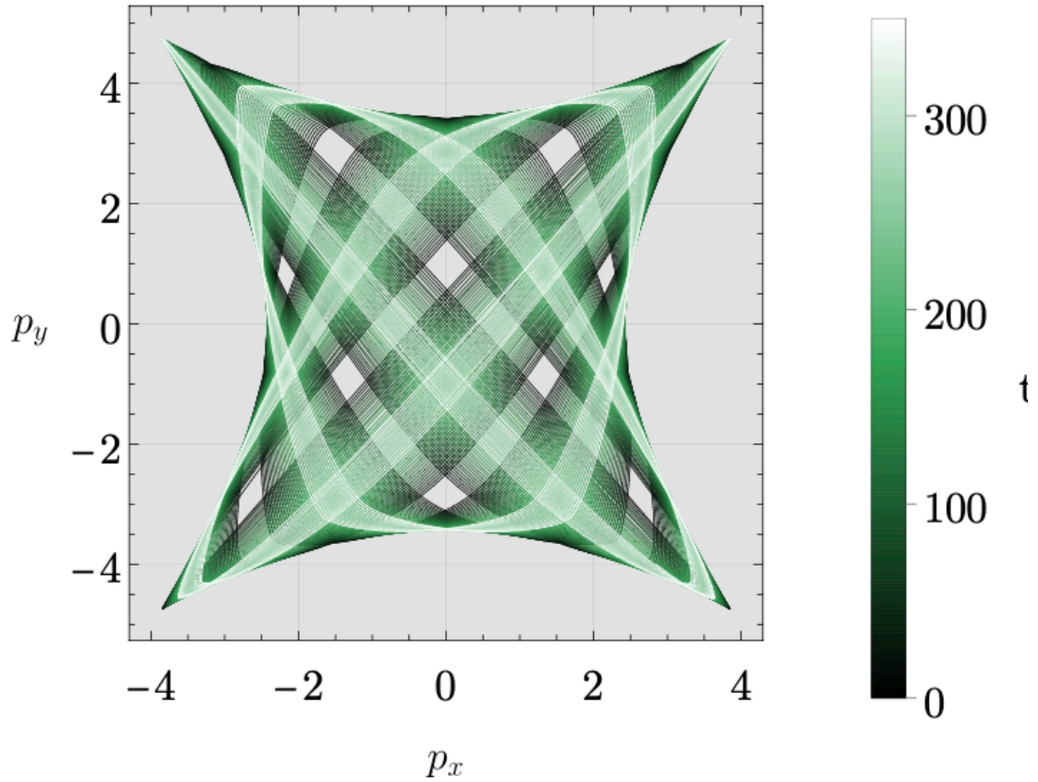
Origin is the only minimum $2\gamma c^2 \geq \left|1 - \omega^2\right|$

$$H_\uparrow = p_x^2 + x^2 + p_y^2 + \omega^2 y^2 + \frac{2}{c}K^2 + \left(V_\uparrow - x^2 - \omega^2 y^2\right) = \text{const} > 0$$

positive definite

$$\omega^2 = 1\,, \gamma = 1\,, c = 1$$





Split in Positive and Negative Energies

$$H = H_{\uparrow} - H_{\downarrow}$$

$$H_{\downarrow} = \frac{1}{2}p_x^2 + \frac{3}{2}p_y^2 + \frac{2}{c}K^2 + V_{\downarrow}(x,y) \qquad K = x\,p_y + y\,p_x$$

$$V_{\downarrow} = \frac{1}{2}x^2 + \frac{3}{2}\omega^2 y^2 + \frac{1}{2c}\left(1-\omega^2\right)\left(x^2-y^2\right)\left(x^2+3y^2\right) + \gamma\left(\left(x^2-y^2\right)^2\left(x^2+3y^2\right) + c\left(x^4+3y^4\right)\right)$$

$V_{\uparrow\downarrow}$ and $H_{\uparrow\downarrow}$ are positive definite

at infinity $V_{\uparrow\downarrow}$ grow without bound

Poisson Brackets $\left\{H_{\uparrow}, H_{\downarrow}\right\} = 0$

Standard Canonical Quantization

$$[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = i$$

$$[\hat{x}, \hat{y}] = [\hat{x}, \hat{p}_y] = [\hat{y}, \hat{p}_x] = [\hat{p}_x, \hat{p}_y] = 0$$

Schrödinger representation

$$\langle \Psi \,|\, \Phi \rangle \equiv \int_{\mathbb{R}^2} dx dy \, \overline{\Psi}\left(x, y\right) \Phi\left(x, y\right)$$

$$\hat{x} = x\,, \qquad \hat{p}_x = -\,i\partial_x\,, \qquad \hat{y} = y\,, \qquad \hat{p}_y = -\,i\partial_y$$

Time Evolution

Standard Schrödinger eq. $$i\frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

Hamiltonian is a *hyperbolic* operator…

$$\hat{H} = -\frac{1}{2}\left(\partial_x^2 - \partial_y^2\right) + \frac{1}{2}\left(x^2 - \omega^2 y^2\right) + V_I(x,y)$$

Is evolution well defined?

Naively, stationary $\hat{H}|\Psi\rangle = E|\Psi\rangle$ solutions look similar to waves and are not normalisable!

Elliptic Split

$$\hat{H}_{\uparrow} = -\,\partial_i \left(M_{\uparrow}^{ik} \partial_k \right) + V_{\uparrow}\left(x, y\right)$$

$$\hat{H}_{\downarrow} = -\,\partial_i \left(M_{\downarrow}^{ik} \partial_k \right) + V_{\downarrow}\left(x, y\right)$$

$$[\hat{H}_{\uparrow}, \hat{H}_{\downarrow}] = 0$$

$$M_{\uparrow}^{xx} = 1 + \frac{2}{c} y^2, \quad M_{\uparrow}^{yy} = 1 + \frac{2}{c} x^2, \quad M_{\uparrow}^{xy} = \frac{2}{c} xy$$

Both integrals of motion are positive-definite *elliptic* self-adjoint/Hermitian operators!

$\hat{H}_{\uparrow\downarrow}$ **both have only *positive discrete* eigenvalues** $E_n^{\uparrow}$ **and** $E_m^{\downarrow}$

$\hat{H}_{\uparrow\downarrow}$ **and** $\hat{H}$ **share the complete system of eigenvectors** $\left| E_n^{\uparrow}, E_m^{\downarrow} \right\rangle$

$\hat{H}$ **has a pure point spectrum** $\hat{H} \left| E_n^{\uparrow}, E_m^{\downarrow} \right\rangle = \left(E_n^{\uparrow} - E_m^{\downarrow} \right) \left| E_n^{\uparrow}, E_m^{\downarrow} \right\rangle$

Vacuum

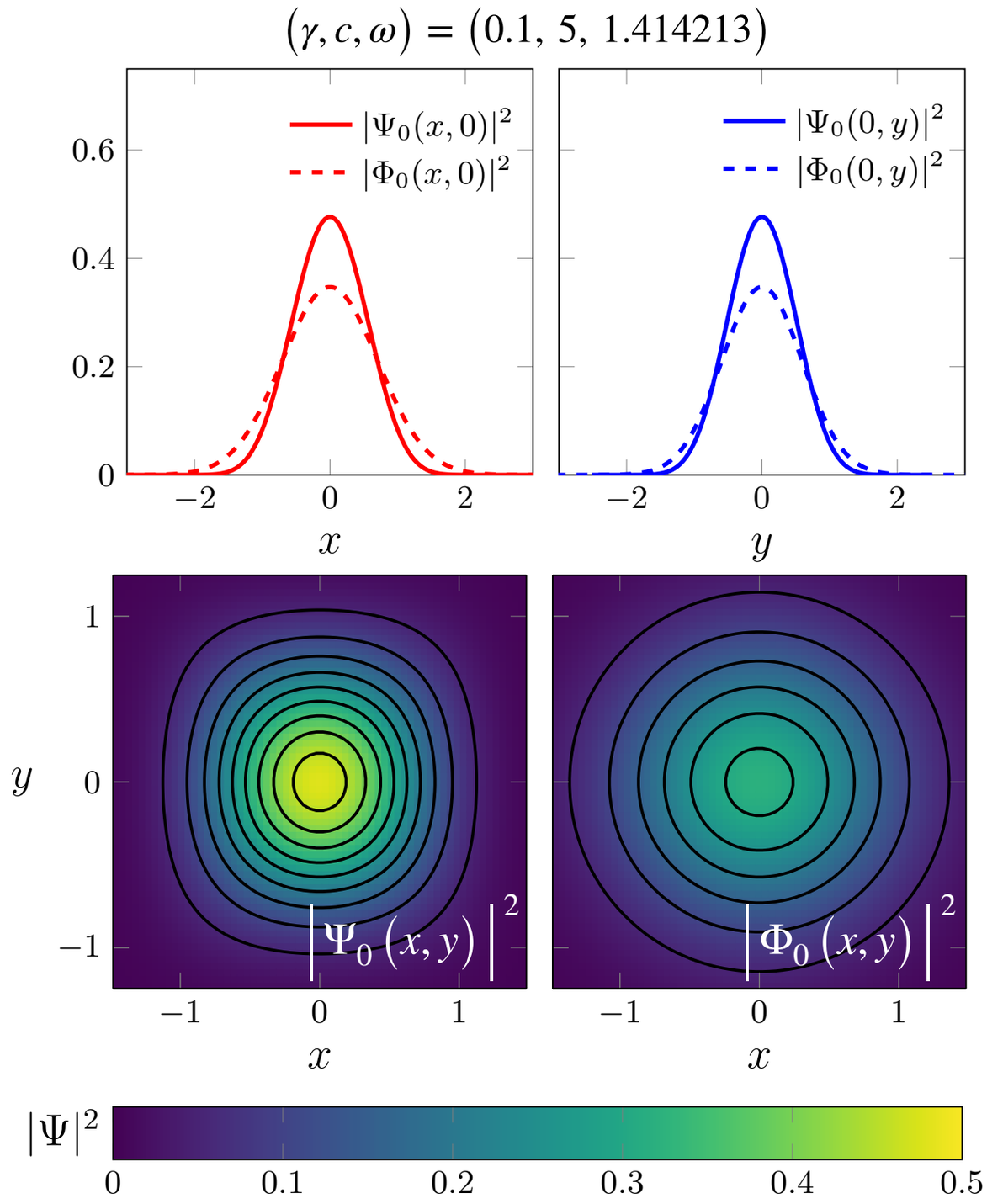
$$|\Psi_0\rangle = |E_0^{\uparrow}, E_0^{\downarrow}\rangle$$

is unique i.e. non degenerate

$$\Psi_0\left(x,y\right) = \langle x, y\,|\, E_0^{\uparrow}, E_0^{\downarrow}\rangle$$

Positive

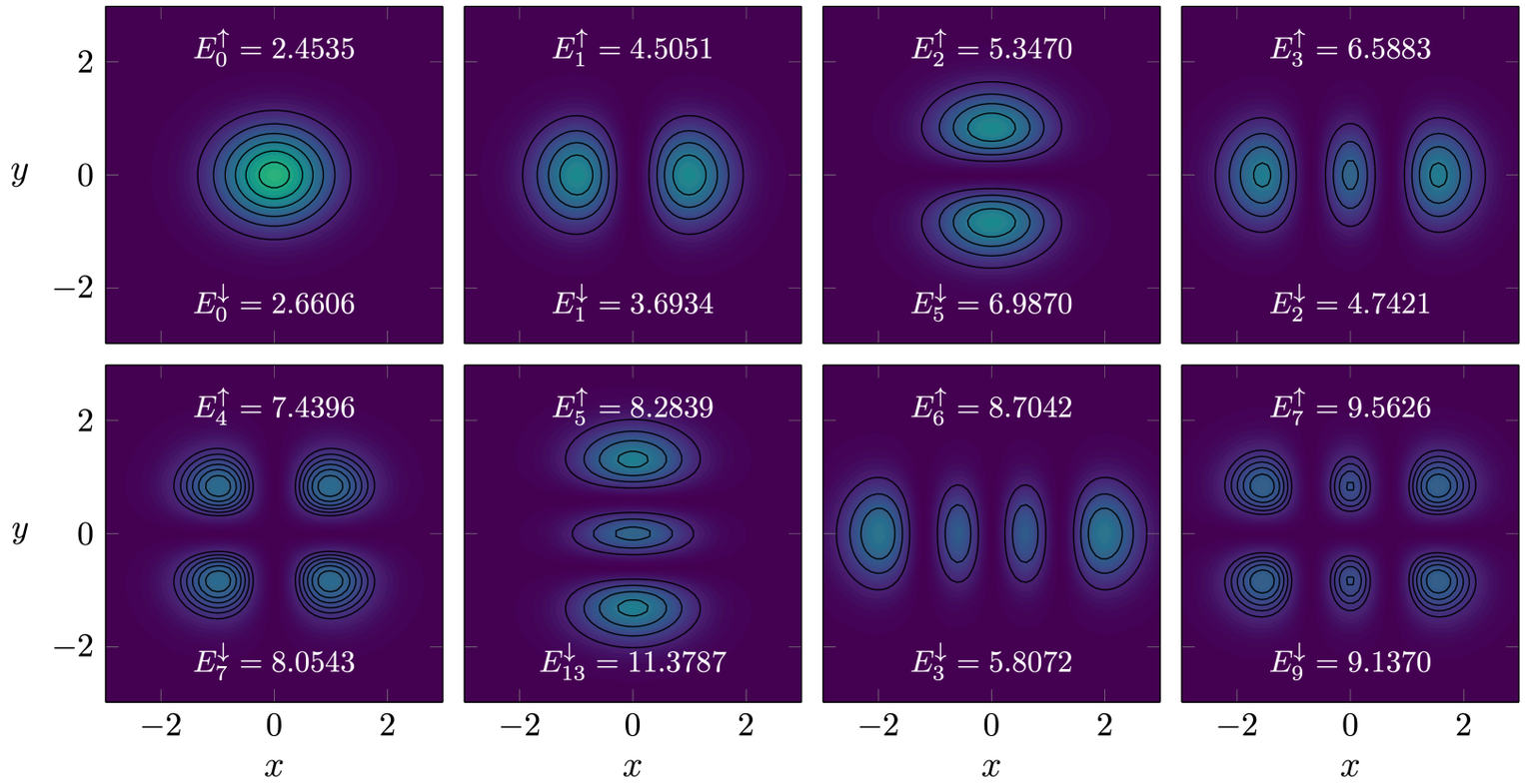
$$\Phi_0\left(x,y\right) = \left(\frac{\omega}{\pi^2}\right)^{\frac{1}{4}} \exp\left(-\frac{x^2+\omega y^2}{2}\right)$$



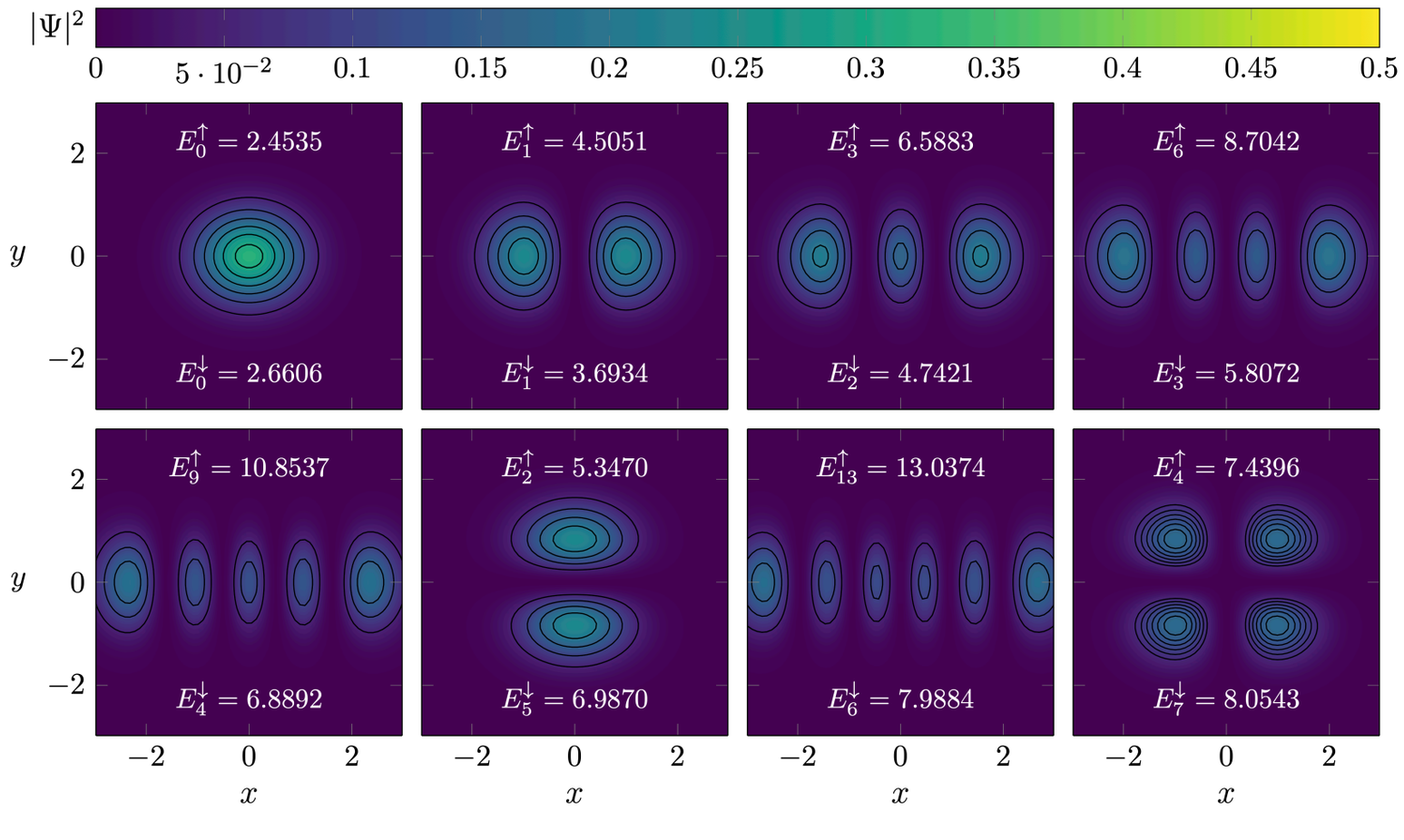
Interacting vacuum with ghost $\Psi_0\left(x,y\right)$ is more confining than decoupled one $\Phi_0\left(x,y\right)$!

First 8 States, for weakly coupled case $\omega = 1.414213$, $c = 10^2$, $\gamma = 10^{-4}$

Ordered by $\hat{H}_{\uparrow}$

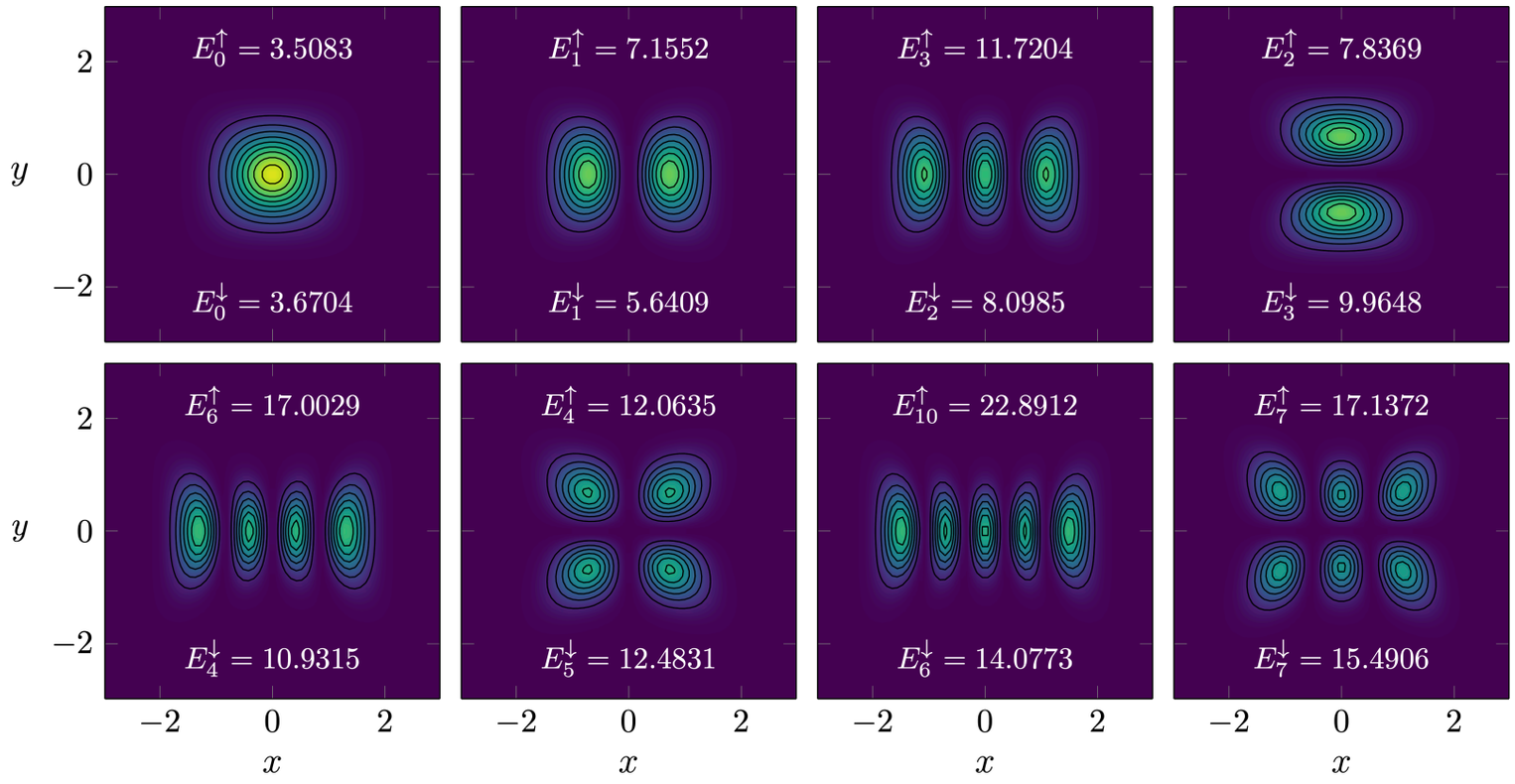


Ordered by $\hat{H}_{\downarrow}$



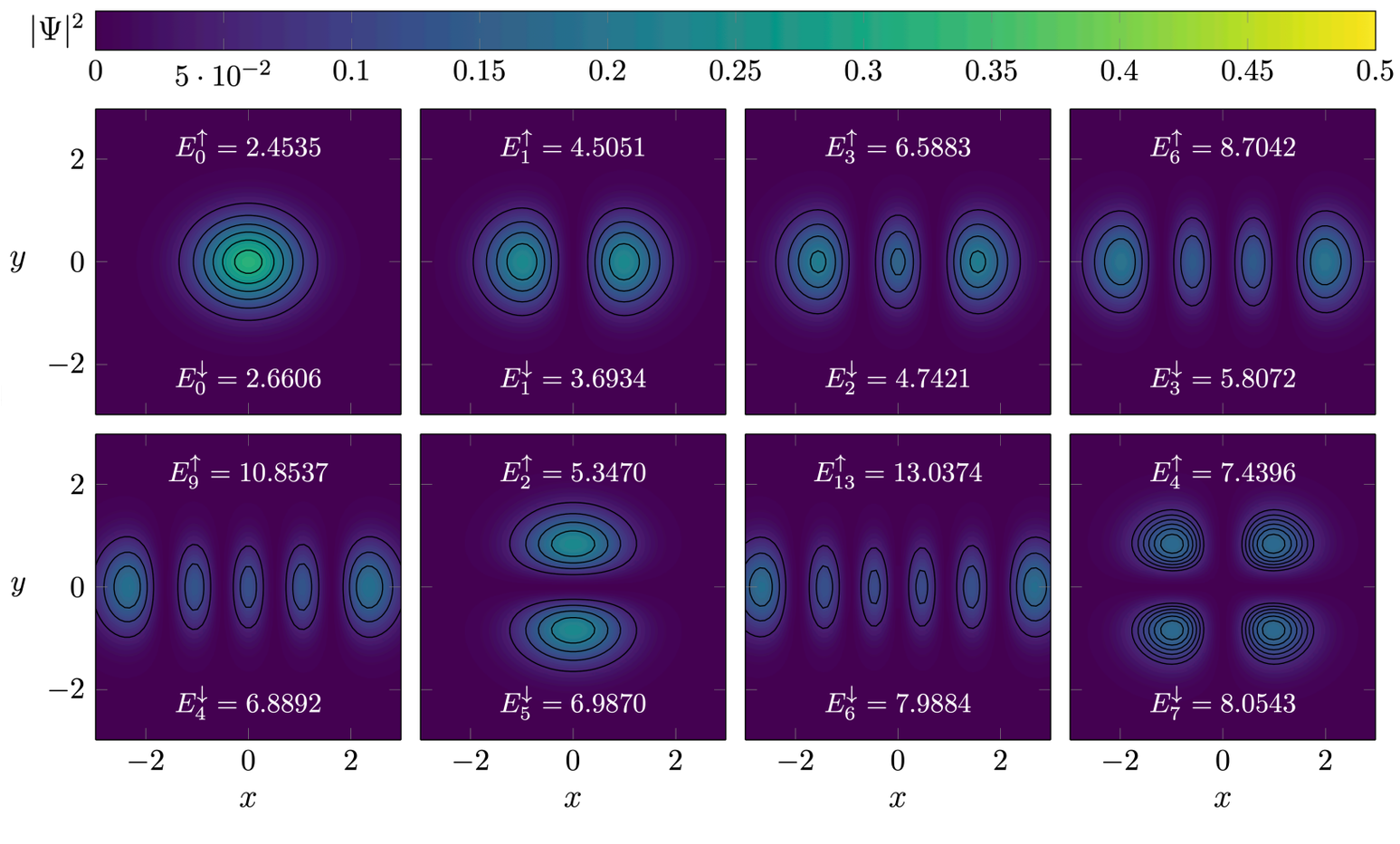
First 8 States, $\omega = 1.414213$, strongly coupled vs weakly coupled case

Strong Coupling
$c = 5, \gamma = 0.1$
Ordered by $\hat{H}_\downarrow$

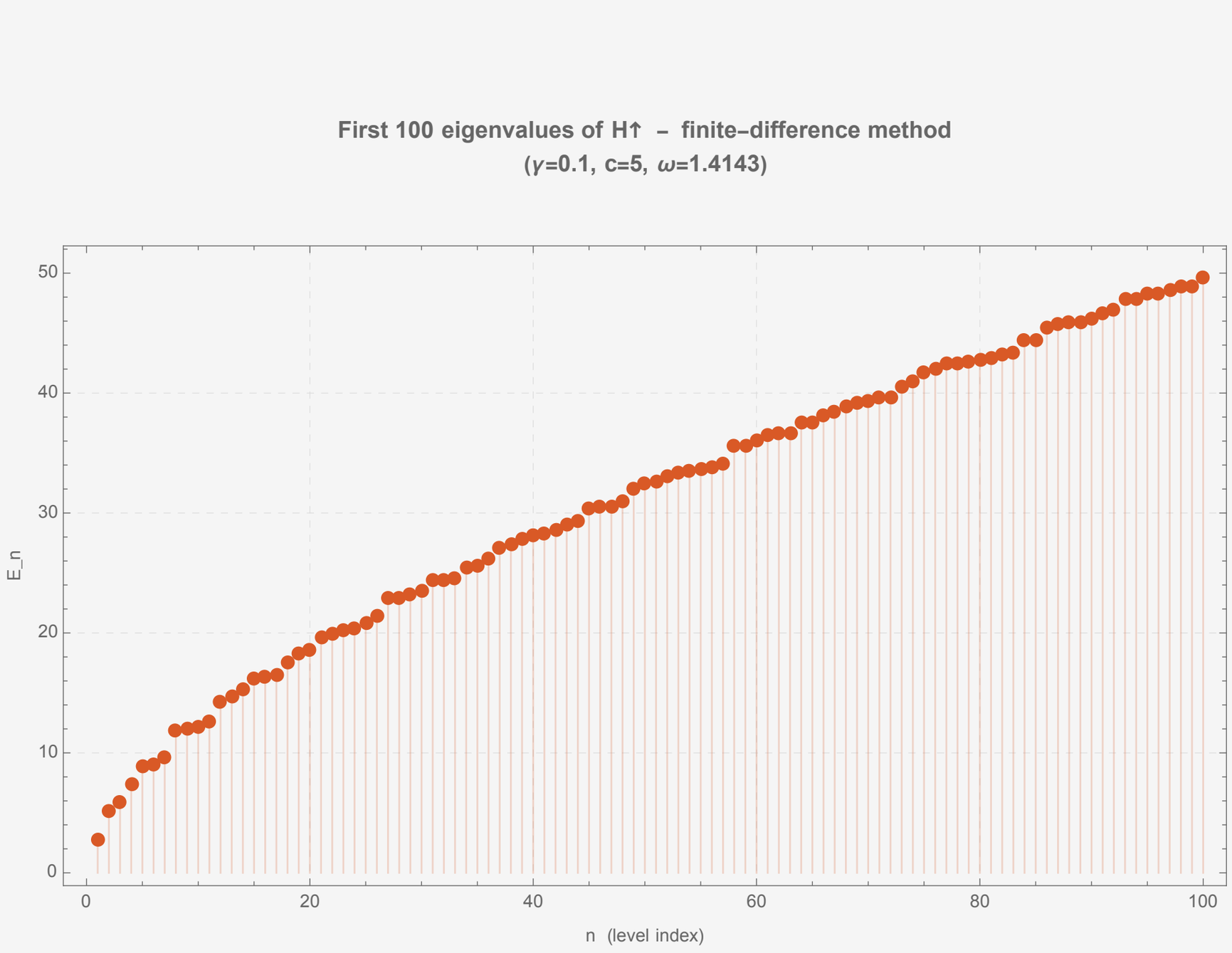


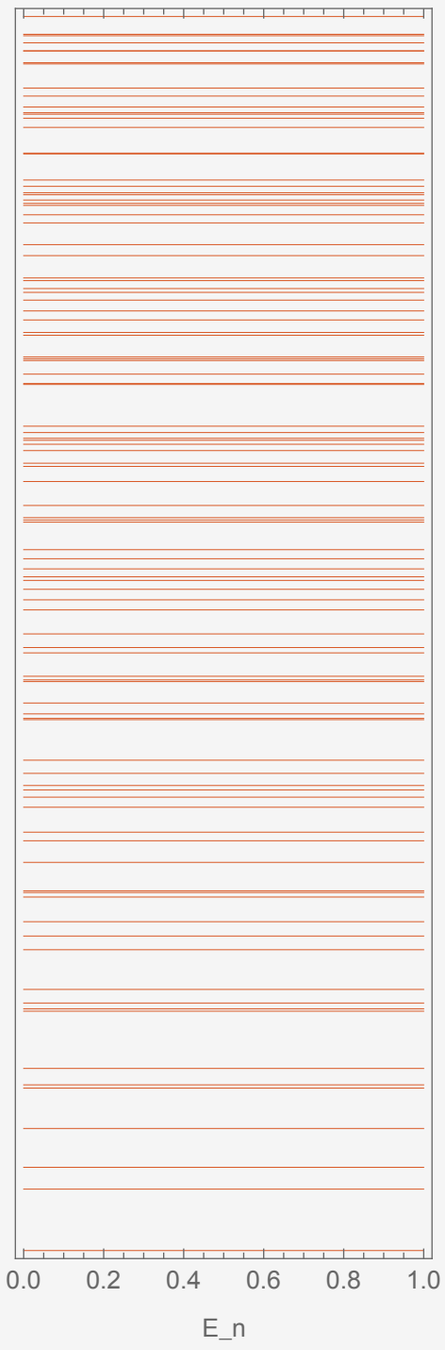
Confined more!

Weak Coupling
$c = 10^2, \gamma = 10^{-4}$
Ordered by $\hat{H}_\downarrow$



$\hat{H}_{\uparrow}$ Spectrum





"Back in the Future"

$$[\hat{H}_{\uparrow}, \hat{H}_{\downarrow}] = 0$$

Manifestly unitary operators

$$\hat{U}_{\uparrow}(t) = e^{-it\hat{H}_{\uparrow}} \qquad \hat{U}_{\downarrow}(t) = e^{it\hat{H}_{\downarrow}}$$

Manifestly Unitary Evolution!

$$\hat{U}(t) = e^{-it\hat{H}} = \hat{U}_{\downarrow}(t)\,\hat{U}_{\uparrow}(t) = e^{it\hat{H}_{\downarrow}} e^{-it\hat{H}_{\uparrow}} = e^{-it\hat{H}_{\uparrow}} e^{it\hat{H}_{\downarrow}}$$

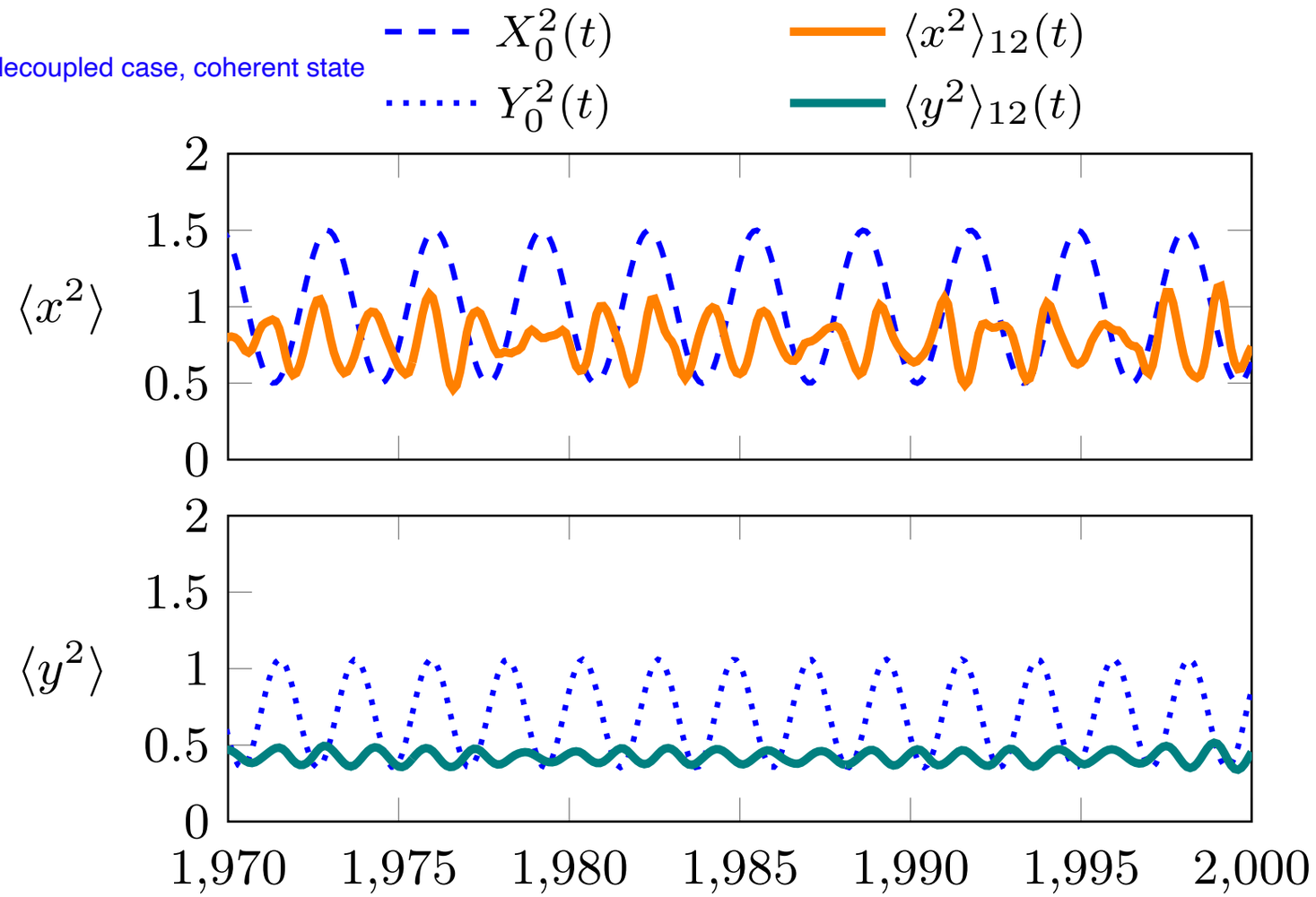
“Quantum Bound on Motion”

for an arbitrary state $|\Psi\rangle$ **their sum is the smallest in vacuum $|\Psi_0\rangle$!!!**

$$\langle \hat{H}_{\uparrow} \rangle = \langle \hat{p}_x^2 \rangle + \langle x^2 \rangle + \langle \hat{p}_y^2 \rangle + \omega^2 \langle y^2 \rangle + \frac{2}{c} \langle \hat{K}^2 \rangle + \langle \left(V_{\uparrow} - x^2 - \omega^2 y^2 \right) \rangle = \text{const} > 0$$

$$\Psi_G(x,y) = \left(\frac{\omega}{\pi^2} \right)^{\frac{1}{4}} \exp\left(-\frac{1}{2}(x - x_0)^2 - \frac{1}{2}\omega(y - y_0)^2 + i\left(x p_{0x} + y p_{0y} \right) \right)$$

for decoupled case, coherent state



Fluctuations are more confined in interacting ghost case than in the case of two oscillators!

Evolution of the Gaussian, “decoupled” case

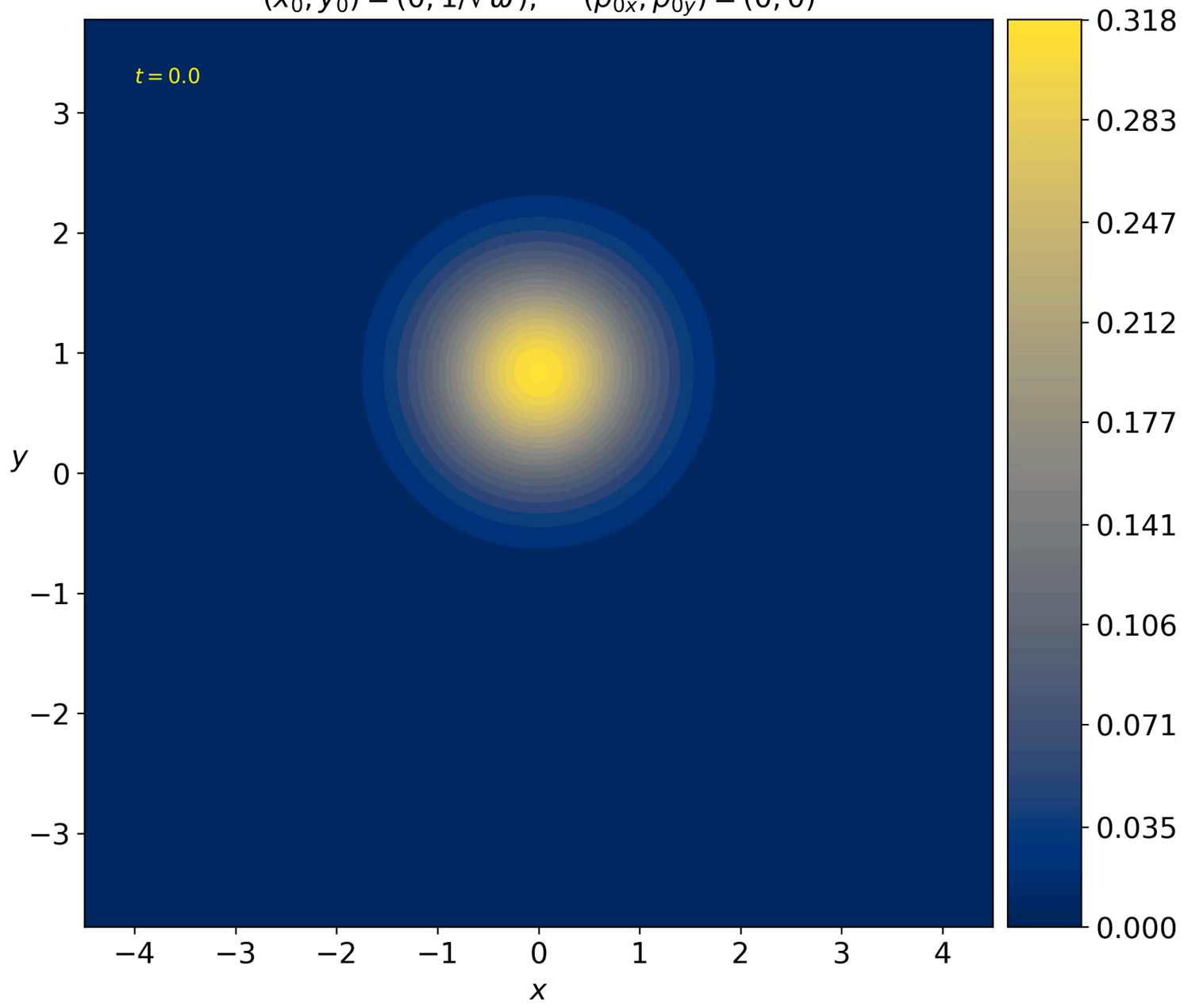
https://doi.org/10.5281/zenodo.19710447

$$\Psi_G(x,y) = \left(\frac{\omega}{\pi^2}\right)^{\frac{1}{4}} e^{-\frac{1}{2}(x-x_0)^2 - \frac{1}{2}\omega(y-y_0)^2 + i\left(xp_{0x} + yp_{0y}\right)}$$

Animation of $|\Psi(t)|^2$, almost decoupled

$c = 1e+10, \quad \gamma = 1e-20, \quad \omega = 1.414213$

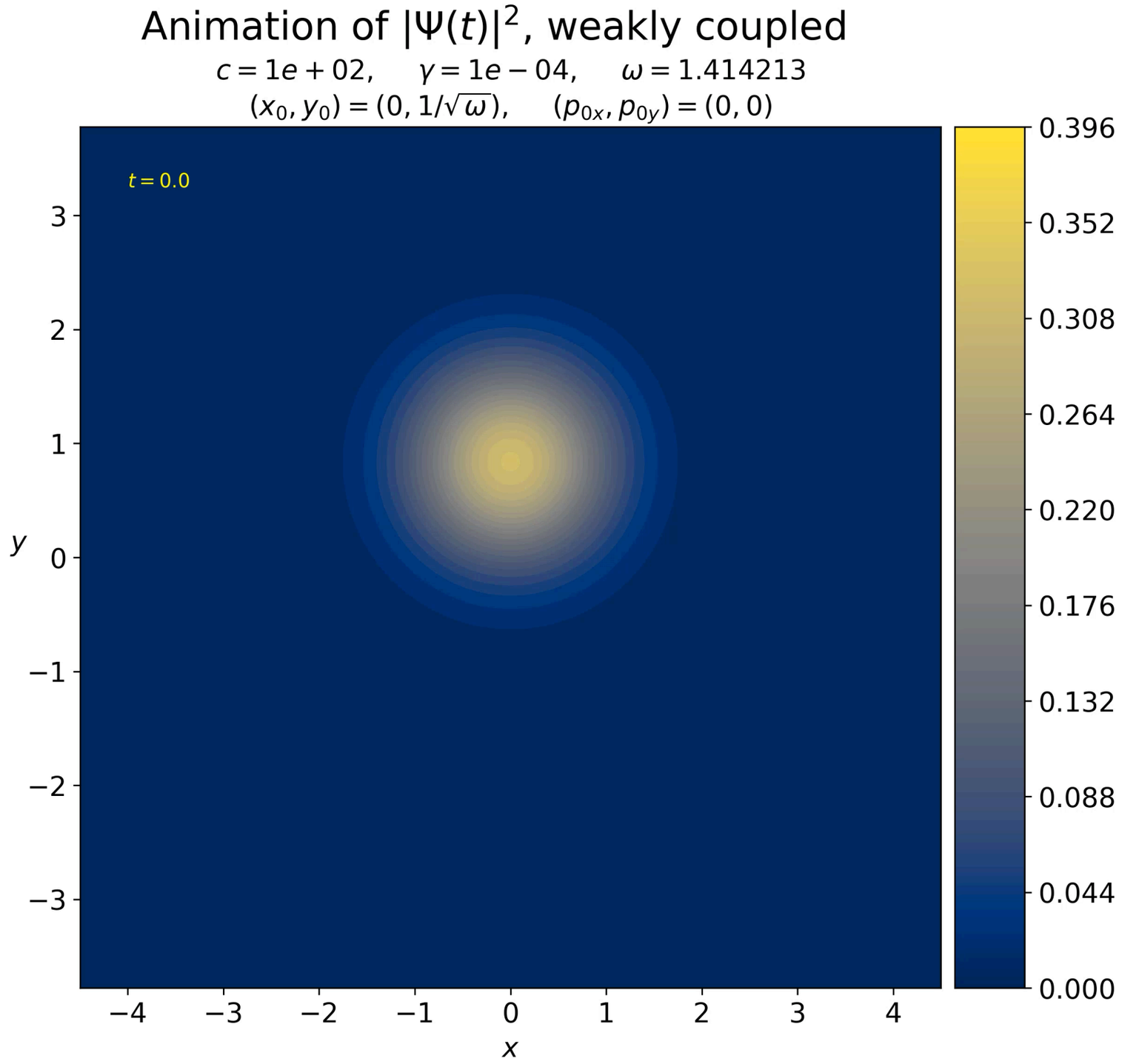
$(x_0, y_0) = (0, 1/\sqrt{\omega}), \quad (p_{0x}, p_{0y}) = (0, 0)$



Evolution of the Gaussian, weakly coupled case

https://doi.org/10.5281/zenodo.19710447

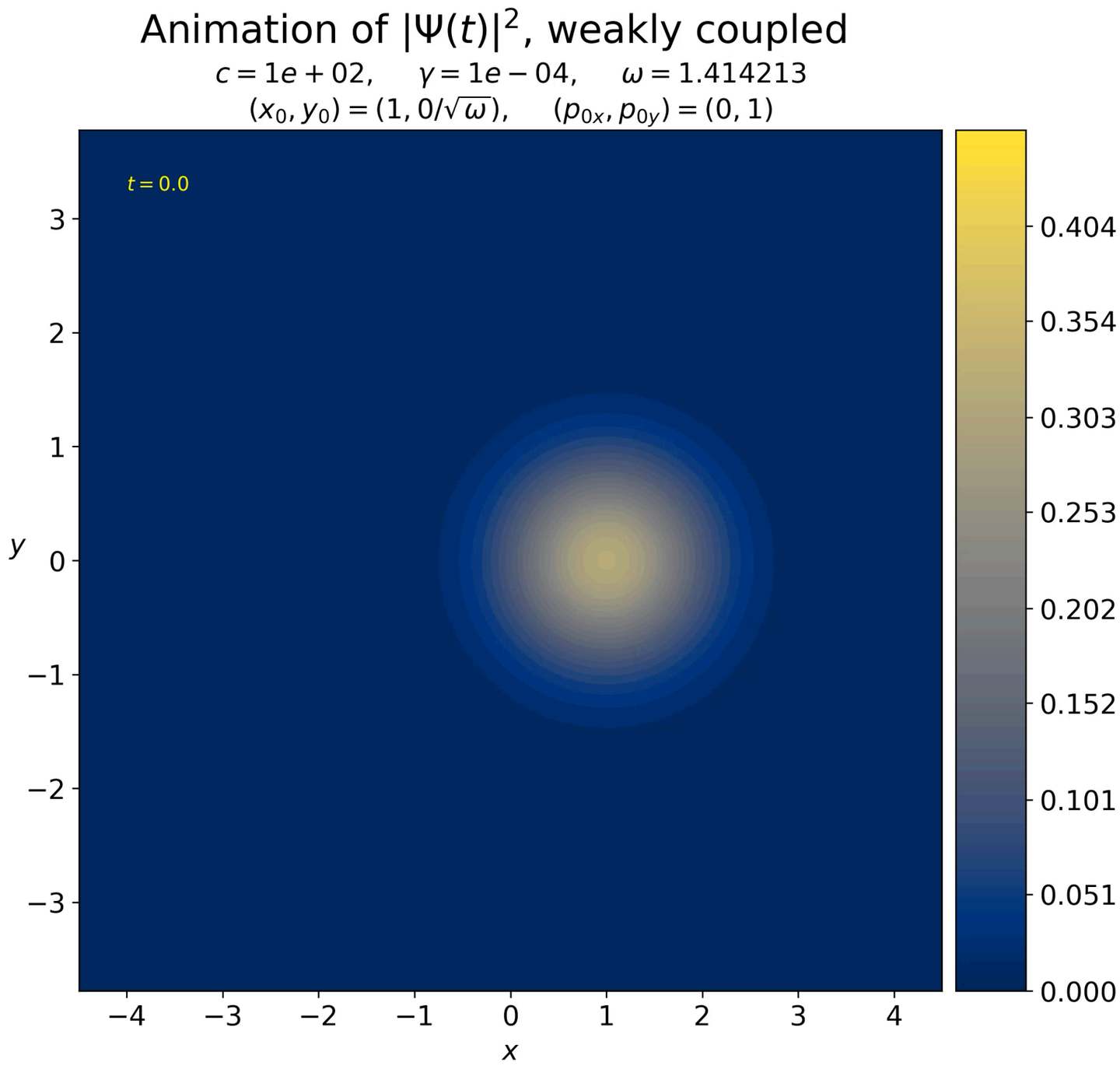
$$\Psi_G(x,y) = \left(\frac{\omega}{\pi^2}\right)^{\frac{1}{4}} e^{-\frac{1}{2}(x-x_0)^2 - \frac{1}{2}\omega(y-y_0)^2 + i\left(xp_{0x} + yp_{0y}\right)}$$



Evolution of the Gaussian, weakly coupled case II

https://doi.org/10.5281/zenodo.19710447

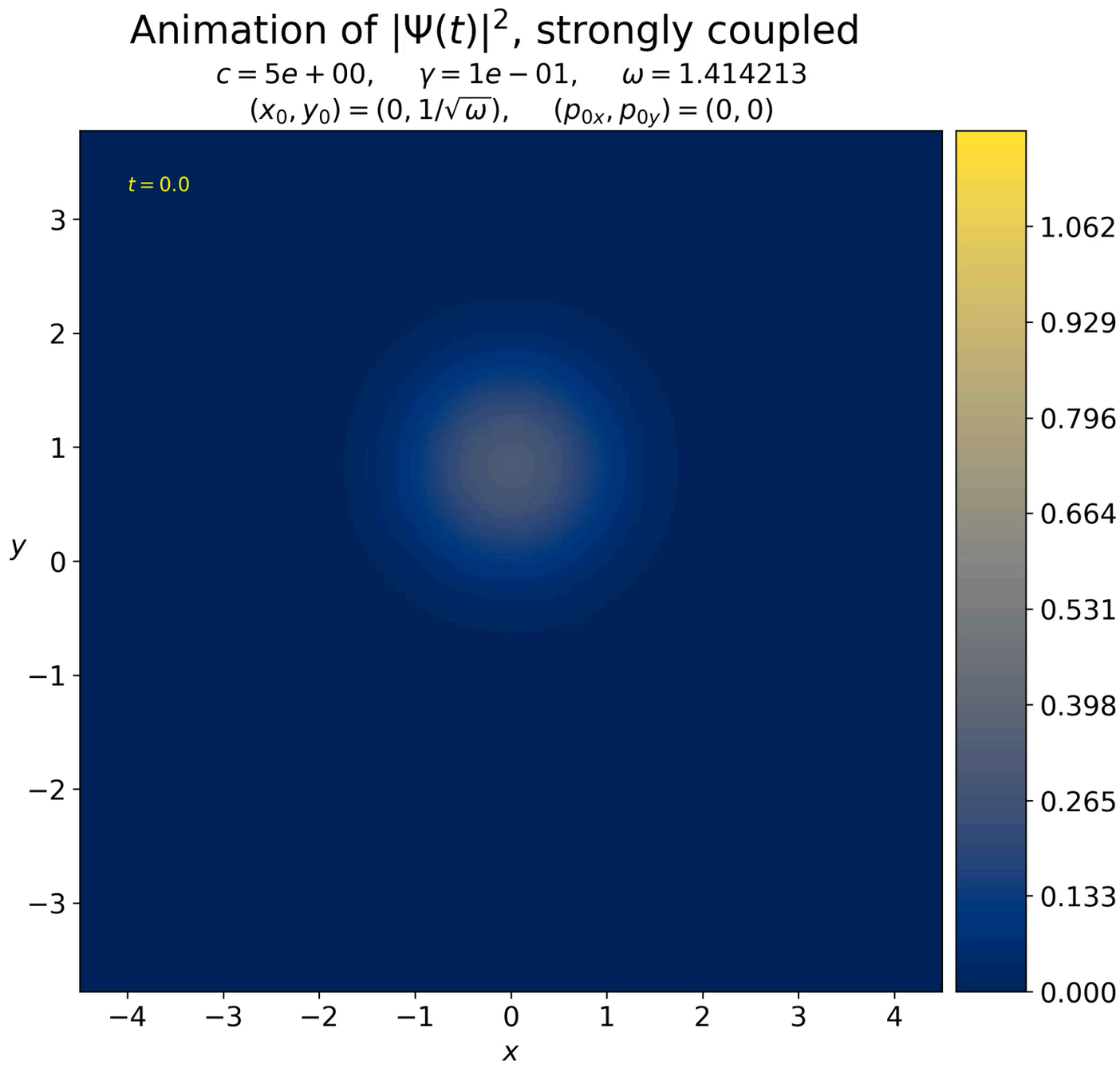
$$\Psi_{\mathrm{G}}(x,y) = \left(\frac{\omega}{\pi^2}\right)^{\frac{1}{4}} e^{-\frac{1}{2}(x-x_0)^2-\frac{1}{2}\omega(y-y_0)^2+i\left(xp_{0x}+yp_{0y}\right)}$$

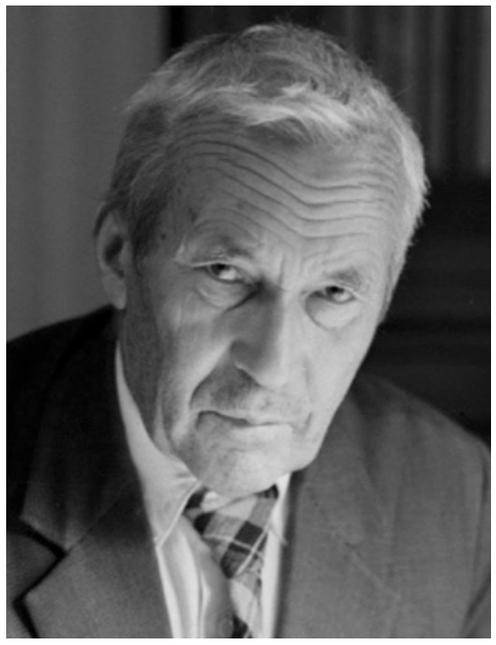
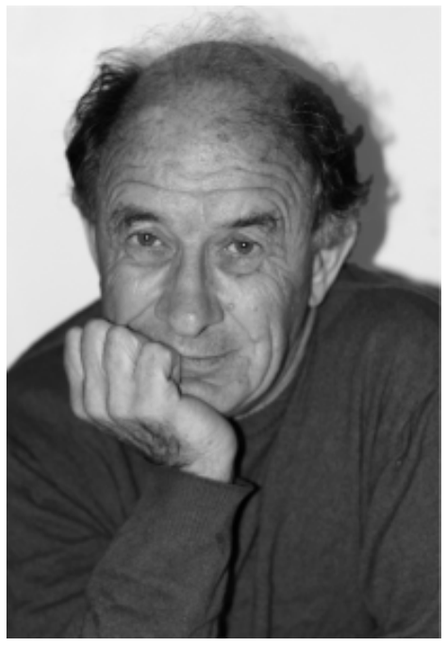
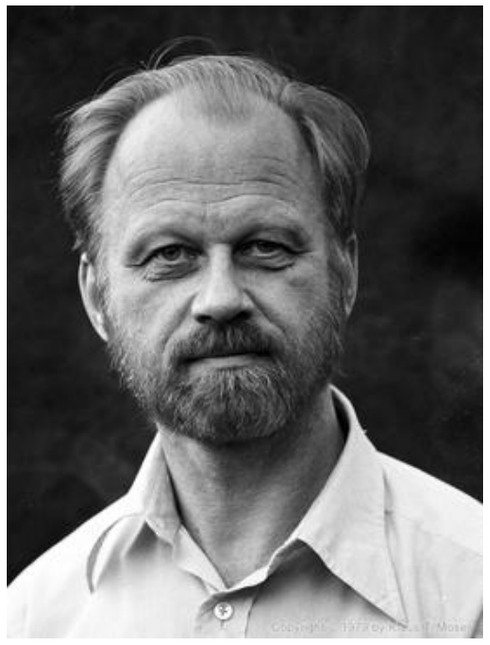


Evolution of the Gaussian, strongly coupled case

https://doi.org/10.5281/zenodo.19710447

$$\Psi_{\mathrm{G}}(x,y) = \left(\frac{\omega}{\pi^2}\right)^{\frac{1}{4}} e^{-\frac{1}{2}(x-x_0)^2-\frac{1}{2}\omega(y-y_0)^2+i\left(xp_{0x}+yp_{0y}\right)}$$



Kolmogorov–Arnold–Moser (KAM) theorem

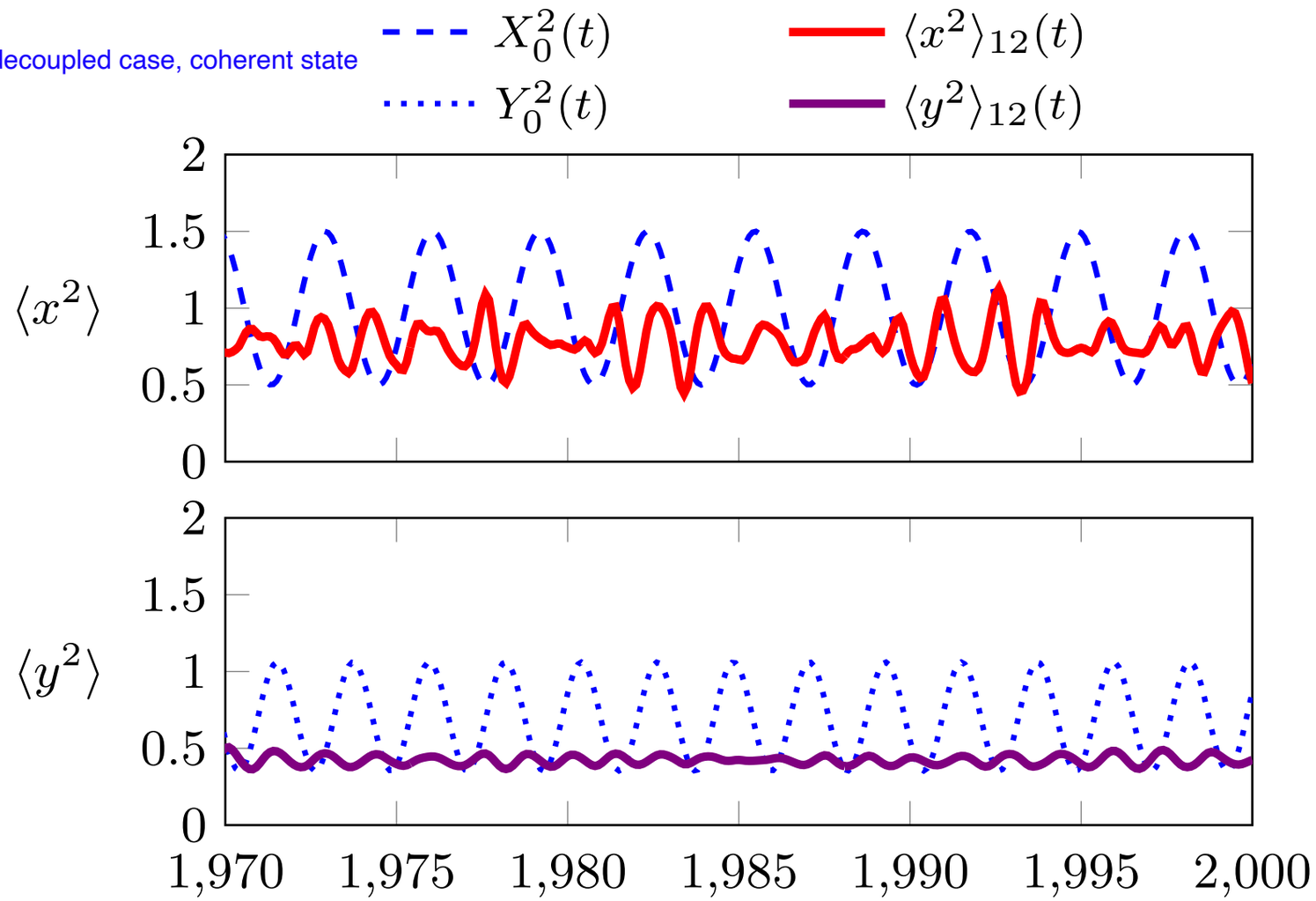
Small *structural* changes
do not jeopardise
the stability and finiteness
of motion

Non-Conservation for $\delta\hat{H}_I$ is Slow

$$\frac{d}{dt}\langle \hat{H}_{\uparrow}(t)\rangle = i\langle[\delta\hat{H}_I, \hat{H}_{\uparrow}(t)]\rangle$$

for $\delta\hat{H}_I = \alpha\, x^2\, y^2$ with $\alpha = 0.01$

for decoupled case, coherent state



Maybe they are not always hostile...
Thanks a lot for attention!